Prabhendra Singh

LinkedIn | GitHub |Hackerrank|(+91)6377290428| prabhendras@gmail.com

Summary

Data Scientist and **Generative AI Engineer** with over **3.8 years** of experience designing and **deploying** LLM- driven solutions in healthcare and analytics domains. Proven expertise in Retrieval-Augmented Generation **(RAG)**, **Prompt Engineering**. Adept in **Python**, **NLP**, **Databricks**, **FastAPI**. Currently contributing to AI innovation at Harita Hive, leveraging LLMs for intelligent healthcare applications.

Strong academic foundation with a **Master's** in **Geoinformatics** from **Central University of Karnataka**.

Technical Skills

Programming	Python
Database	SQL Server, MySQL
AI/ML & Analytics	ML, NLP, Gen AI, , Agentic AI, RAG, Prompt Engineering, Data Visualization
Frameworks & Tools	LangChain, Autogen, KnowledgeGraph, Scikit-learn, Vector Databases, Power BI, MLflow, FastAPI, Streamlit, XGBoost, Git, OpenAI, Hugging Face Transformers, BERT,CNN
Cloud & DevOps	Databricks, Azure, Azure App Service, Docker

Professional Experience

Harita Hive Private Limited ***Duration: 1 May 2023 – Till Date***

Harita Hive is a specialized provider of healthcare technology and business process services to healthcare technology companies, healthcare providers, managed care organizations, health plans, and disease management companies.

Project	**Gen AI Assistant using FHIR Data** **Duration: Sept 2024 – Present**
Summary	Developed a GenAI-powered search engine to assist payers in querying FHIR databases with simplified queries, enhancing accuracy and providing advanced analytics
Main Technologies	Python, FastAPI, LLMs (OpenAI / Azure OpenAI), LangChain, Azure Data Lake, Prompt Engineering, Docker
Responsibilities	▪ Built a GenAI-powered natural language query engine for FHIR datasets, enabling non-technical users to extract insights effortlessly. ▪ Integrated LangChain-based RAG pipeline to generate contextual, accurate clinical responses. ▪ Developed secure REST APIs for accessing key resources (Claims, Medications, etc.) used by payer applications. ▪ Logged and tracked experiments with MLflow and deployed services via Azure App Service for secure access. ▪ Containerized application components using Docker to ensure consistent and portable deployment across environments.

Project	Prior Authorization on Azure Databricks Duration: Jan 2024 – Sept 2024
Summary	Built a Gen AI-enabled solution to automate and optimize prior authorization workflows in healthcare. Leveraged Databricks for scalable ML processing and integrated RAG-based Gen AI to enhance document understanding and decision support.
Main Technologies	Python, FastAPI, Databricks, MLflow, Rag architecture, Prompt engineering, FAISS Index ,Docker
Responsibilities	▪ Designed and implemented a scalable Gen AI pipeline using RAG for real-time document understanding in prior authorization workflows. ▪ Created FAISS-based vector search for fast document retrieval from Delta Tables. ▪ Enabled real-time summarization and decision support through REST APIs and Angular-based UI. ▪ Ensured model governance with Unity Catalog and deployed via managed endpoints.

Project	Time Series Analysis and Forecasting Duration: May 2023 – Jan 2024
Summary	This time series project in Python aims to analyze temporal data patterns, make informed predictions, and derive valuable insights from sequential data
Main Technologies	Python, Prophet, Xgboost, Scikit-learn, ARIMA, Matplotlib and Seaborn
Responsibilities	▪ Created a pipeline for collecting and preprocessing time-stamped data using Pandas and NumPy ▪ Conducted Exploratory Data Analysis (EDA) to uncover trends, patterns, and anomalies in historical time series data ▪ Extracted relevant features from the time series dataset, such as lag features and rolling statistics, using Pandas ▪ Chose appropriate models like Prophet, Xgboost, and machine learning-based models from Scikit-learn ▪ Actively participated in model refinement and optimization, incorporating feedback from model performance evaluations

MOVATE TECHNOLOGIES PRIVATE LIMITED ***07-OCT-2022 TO 02-APR-2023***

Movate Technologies Pvt. Ltd. is a technology and analytics services provider supporting autonomous vehicle programs like Zoox (Amazon) through LiDAR data processing, geospatial analysis, and AI/ML–driven automation to enhance perception, safety, and vehicle intelligence systems.

Project	Healthcare Marketing Analytics Optimization Duration: Oct 2022 – April 2023
Summary	To automate competitor research and extract digital presence metrics of clinics, hospitals, and healthcare professionals.
Main Technologies	Python, BeautifulSoup, Selenium, Pandas, Regex, Google Search API
Responsibilities	▪ Developed web scraping pipelines to extract SEO keywords, Google reviews, ratings, and service offerings from competitors' websites. ▪ Parsed structured and unstructured data from platforms like Google Maps, JustDial, Practo. ▪ Analyzed extracted data to benchmark client performance against competitors.
	▪ Provided insights on patient reviews and ratings to improve brand reputation strategies. ▪ Automated weekly data refresh and integration into campaign performance reports. ▪ Automated manual Excel analysis for digital marketing data by building a Python script that processes CSV files and generates campaign performance reports. ▪ Saved 80% time for the team and improved accuracy of insights like CTR, CPL, and keyword trends.

Educational Background

- **Internship - GeoSpatial Data Science,** National Institute of Rural Development, Year Oct 2021 - May 2022
- **M.Sc – Geoinformatics**, Central University, Karnataka, Year 2020 - 2022
- **B.Sc – Applied Geoinformatics & Geography (Hons)**, Central University, Karnataka, Year 2017 - 2020

Certifications

- **Machine Learning A-Z™: Hands-On Python In Data Science** (Udemy)
- **Databricks Certified Data Engineer Associate** (Databricks)
- **NLP: Natural Language Processing with Python** (Udemy)
- **LangChain: Develop LLM Powered applications with LangChain** (Udemy)
- **ChatGPT Masterclass: Build Solutions and Apps with ChatGPT** (Udemy)
- **PW Skills: Full Stack Data Science**
- **Harita Hive Certified GEN AI Expert-L2**